

Execute simulation experiments according to your final trial plans to arrive at the desired data. Plan a number of simulation runs and prepare for the variation of parameters and models to get reliable results.

- **Analyze the results of the experiments**

Analyze and interpret the results of the simulation experiments. Conduct a sensitivity analysis of the most important parameters, data and results.

- **Author the final documentation of the entire simulation project**

Once you are finished with the simulation project, update the notes you made while modeling to create the final documentation of the entire simulation project. This will help you, when you have to update or extend your simulation model or any of its components. Executing simulation experiments is a cyclical and evolutionary process. You will modify and improve your initial simulation model a number of times as you incorporate new insights from previous simulation runs. Thus you will arrive at your final simulation model after several cycles after continuously changing your initial draft of your simulation model.

Go to [Prepare Data for the Simulation with DataFit](#)

Go to [Run Simulation Experiments](#)

Go to [Modeling the Material Flow](#)

Getting to Know Plant Simulation

Get to Know Plant Simulation

You can familiarize yourself with *Plant Simulation* in a number of ways.

You can:

- [View the Videos on YouTube](#)
- [View the Sample Models](#)
- [Consult the Step-by-Step Help](#)

View the Videos on YouTube

The **Videos**, which you can access on the **Start Page**, provide a quick introduction to some of the most commonly used modeling tasks. We constantly add new videos.



Click **Videos** to open the list of **Videos** we offer on *YouTube*. Then, click the respective video to view it.

If we demonstrate a feature in a video, the respective help topic provides a link to the video on *YouTube* and jumps to the respective time in the video. The link shows the time code after the equal sign, =18 in the example below. Each video opens in a tab of its own.

Video

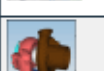
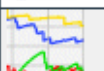
<https://youtu.be/PgT4wkT2Xj8?si=Udlvoq8KcAI9GxQw&t=18>

Note that we use the *YouTube* URL shortener in our links to the videos, for example <https://youtu.be/PgT4wkT2Xj8?si=Udlvoq8KcAI9GxQw&t=18>.

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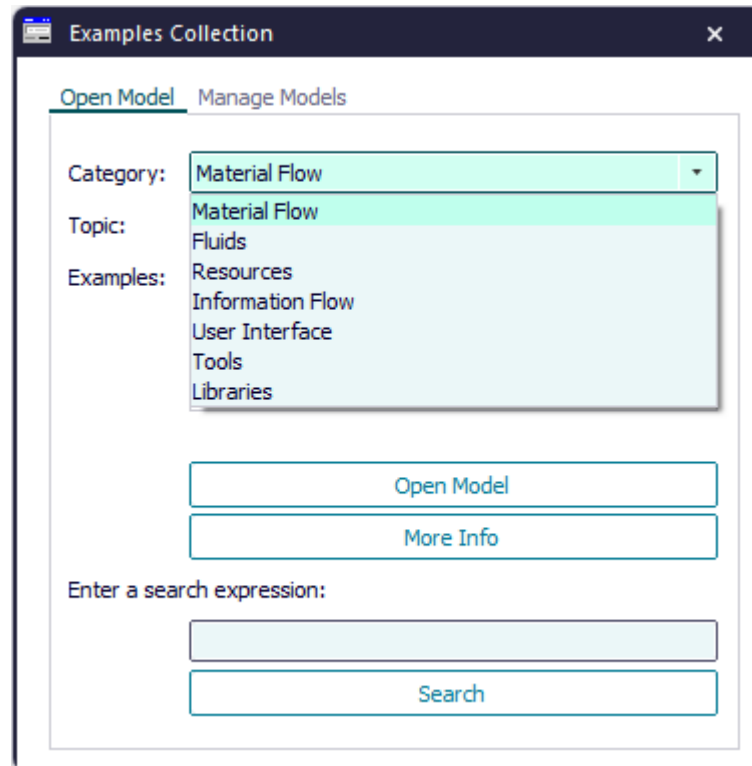
View the Sample Models

To get acquainted with *Plant Simulation*, we recommend to view the sample models. These models demonstrate how to approach a number of problems and will give you ideas on how to solve your modeling tasks.

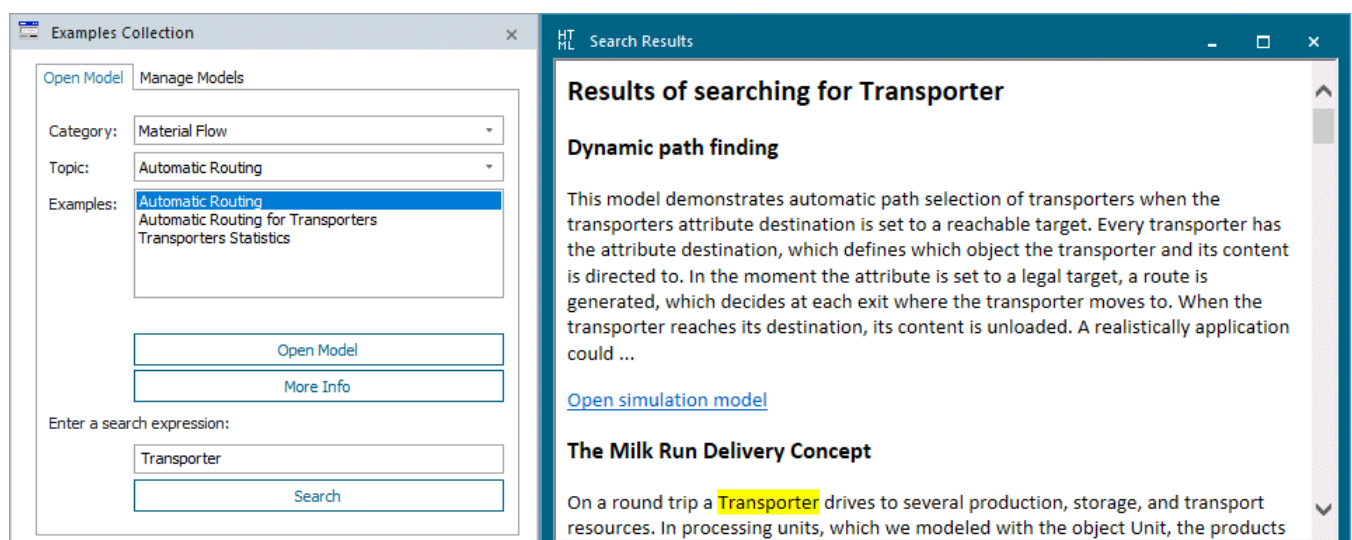
Example Models		
	Factory 51	Shows a concise demonstration model of the 3D visualization.
	Small Parts Production	Shows a simple example of a small parts production and how to use the Experiment Manager.
	Nanobox Assembly	Shows a model of the Nanobox assembly line of Siemens Karlsruhe. A Nanobox is a small computer especially designed for the needs of production facilities.
	Turbo-Charger Assembly	Shows how to increase productivity to meet increasing customer requirements.
	Body in White	Shows how to optimize buffer capacities and how to compare same model with different parameters.
	Rear Axle Assembly	Shows how to optimize the layout and how to define strategies and capacities.
	Python GA Demo	Shows how to optimize a simulation model with genetic algorithms using Python.
	Standard Object Examples	Contains small examples which show how the standard objects can be used.
	Small Examples	Provides sample models demonstrating how to solve various modeling tasks.

To view the sample models, click **Example Models > Small Examples** under **Getting Started** on the **Start Page**.

- Select the respective **Category**, the **Topic**, and the **Sample Model**.



- Click **Open Model** to open the selected.
- Click **More Info** to show a short description of the model.
- Type a search term describing the topic you want to find into the text box **Search** and click **Search** to find a model with this topic. *Plant Simulation* then searches the descriptions in the **ReadMe** files which are part of the models.



If you inadvertently pressed the **Enter** key instead of clicking **Search**, *Plant Simulation* closes the dialog **Examples Collection**. To open the start page again, click the button **Start Page** on the **Window** ribbon tab.

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Consult the Step-by-Step Help

To get acquainted with *Plant Simulation*, we recommend to consult the *Plant Simulation Step-by-Step Help* and the *Plant Simulation Help* that are part of *Plant Simulation*. The former will cover most of your modeling tasks and provide you with information on how to solve them.

Start *Plant Simulation*, go to **File > Help** and select **Contents**. Expand ► [Step-by-Step Help](#) and select the topic you need information about.

If you prefer to read the documentation as a book in consecutive order, consult the PDF file, which you can download from the website.

We recommend to [search the Online Help with Copilot](#).

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Working with Plant Simulation, Basics

Working with Plant Simulation, Basics

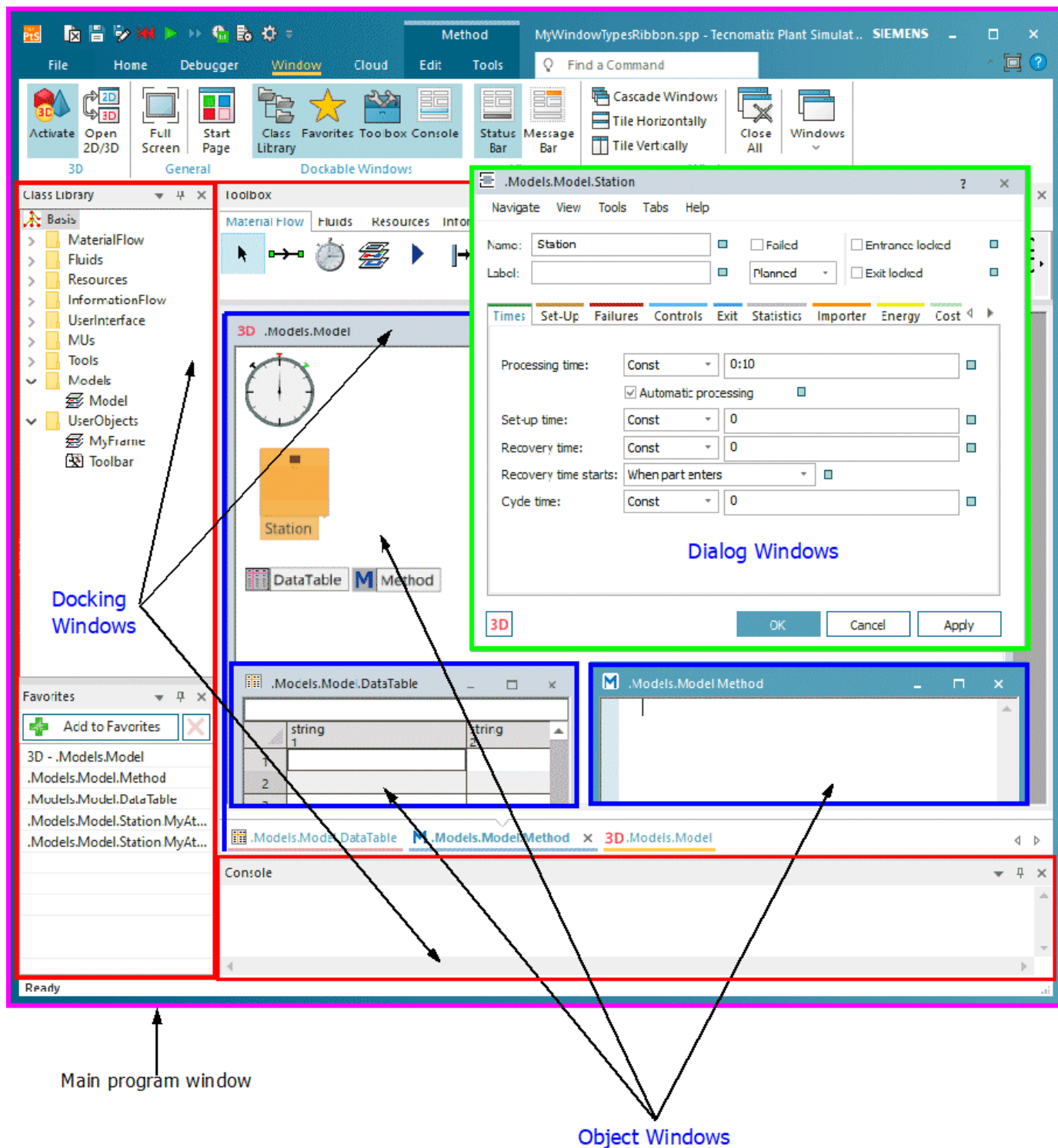
Working with Plant Simulation, Basics provides these topics:

- [Work with Window Types](#)
- [Select Settings in Plant Simulation](#)
- [Select View Options in the Frame](#)
- [Change the Settings of the Objects](#)
- [Find Objects and Text in Your Simulation Model](#)

Working with Window Types

Work with Window Types

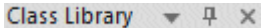
Plant Simulation is a multiple-document interface application (MDI) which shows its windows in the common parent window.



The *Plant Simulation* program window has a magenta border in the figure above.

Note

The different window types open in a certain order within the program window. At times, it may happen that the window you want to work with opens in the background, instead of the foreground, where you would expect it to be. If this is the case, move the windows in the foreground aside to access the window into which you want to enter data.

Instead of closing and reopening the [Class Library](#), the [Favorites](#), the [Toolbox](#), and the [Console](#), you can hide them: Click  **Auto Hide** in the title bar  to auto hide the window. When you then click in another window, *Plant Simulation* hides the window until you move the mouse over the name of the window again. To deactivate auto hide, click .

For further information compare [User Interface Components](#) in the *Reference section* of the *Plant Simulation Help*.

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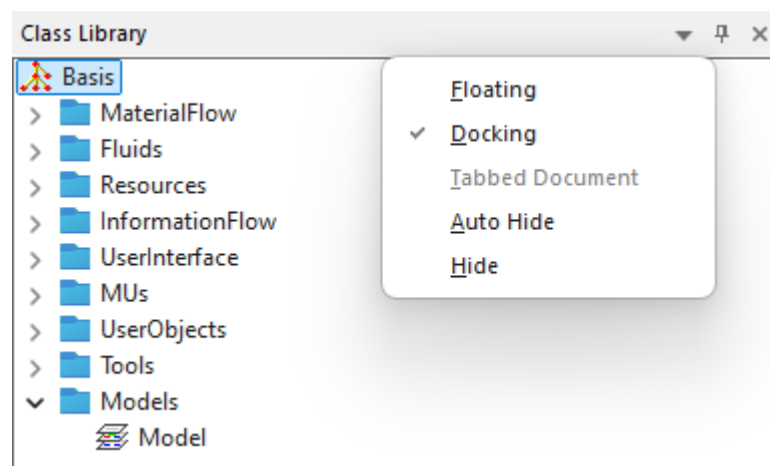
Docking Windows

Docking windows are windows that are docked to a part of the *Plant Simulation* program window. You can undock a docked window and drag it to another part of the window or you can leave it floating freely.

Plant Simulation docking windows are the [Class Library](#), the [Favorites](#), the [Toolbox](#), and the [Console](#).

Docking windows have a red border in the figure below [Work with Window Types](#). *Docking windows* always open in the foreground, on top of any other windows, i.e., they will also be placed on top of any open dialog windows.

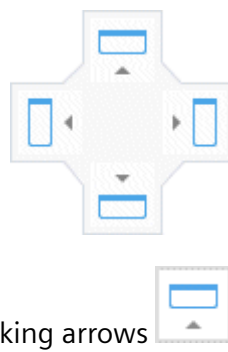
- To undock a docking window, click into the title bar of the window with the right mouse button and select **Floating**.



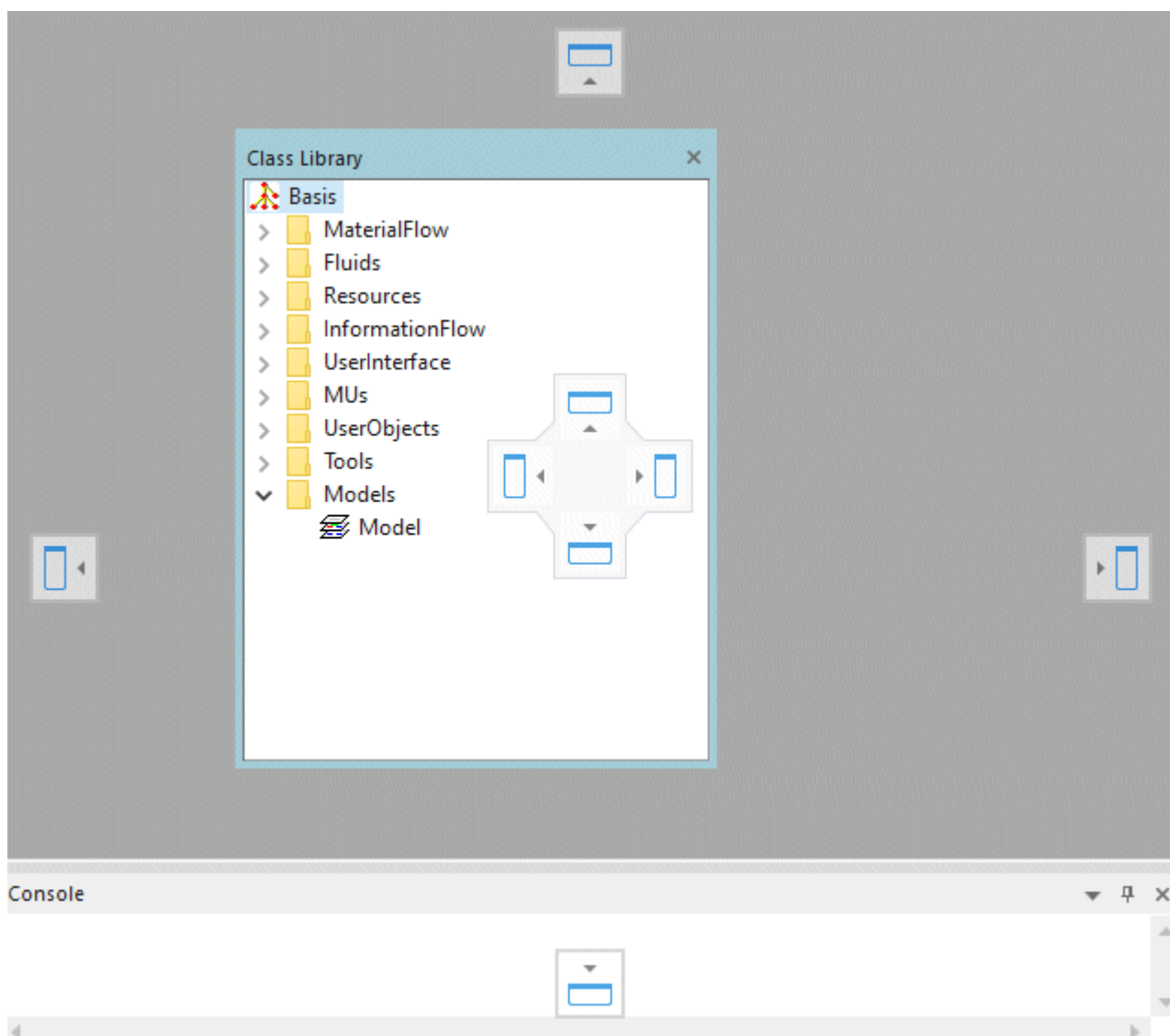
- To redock a floating window again, click in the title bar of the window with the right mouse button and select **Docking**.

Or

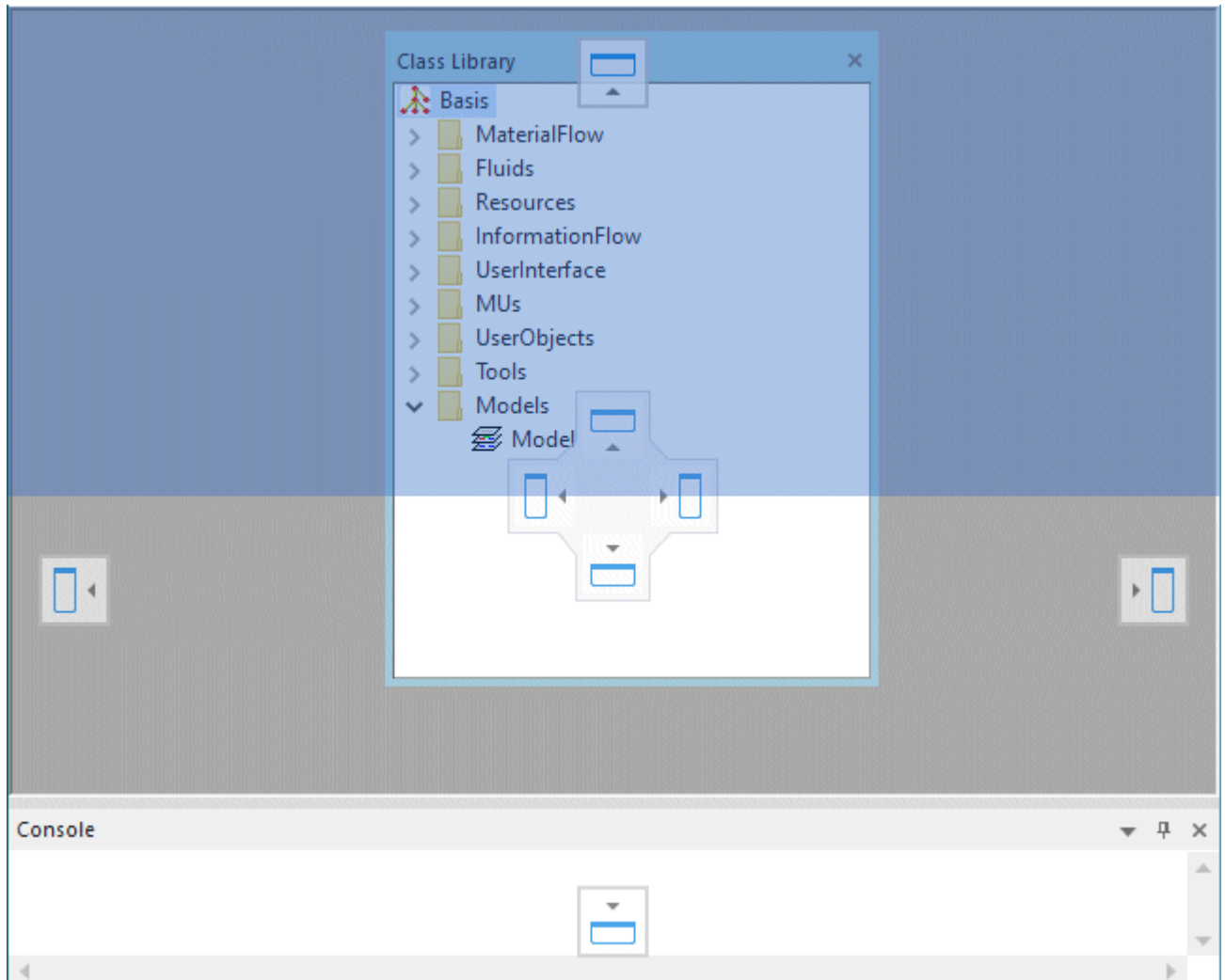
Click the title bar of the docking window and hold down the mouse button. *Plant Simulation* shows a docking indicator pointing to the upper, lower, right, and left border of the program window. Drag the mouse to the docking arrow where you would like to dock the window. *Plant Simulation* highlights this area of the program window in blue. Release the mouse button once the window is at the desired position. *Plant Simulation* then **docks** the window to the respective **docking window**.



Plant Simulation also shows individual docking arrows at the four sides of the parent window, i.e., the *Plant Simulation* program window.



Drag the mouse to the docking arrow of the border of the program window at which you would like to dock the window. *Plant Simulation* highlights this area of the program window in blue. Release the mouse button once the window is at the desired position. *Plant Simulation* then **docks** the window at the respective border of the **parent window**, i.e., the *Plant Simulation* program window.



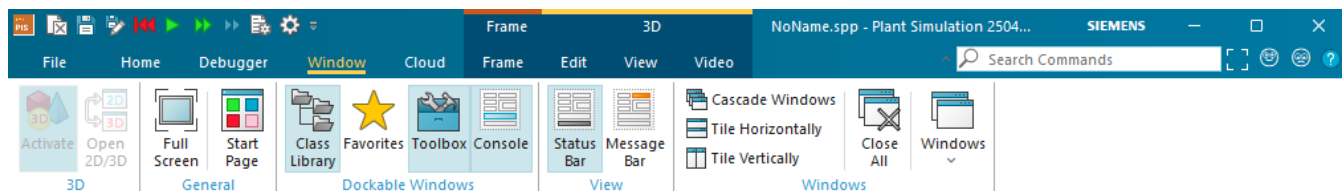
- To dock a floating window to any of the sides of the program window, drag the title bar on a floating window to another location. When you drag the window to the edge of the program window or to a location next to another docked window, it docks there.

Note

When you move a docked window, this might affect the location and size of other windows in the same row within the program window.

- To prevent a floating window from docking when you drag it, hold down **Ctrl**, and move the window to a position of your choice.
- To make a docked window a floating window, double-click the move handle or hold down **Ctrl**, and drag the window.
- To dock a floating window again, double-click its title bar or drag the window to one of the sides of the program window or to a location next to another docked window, it docks there.
- To close a floating window, click **Close**  in the title bar.

- To show or hide the *Class Library*, click  on the **Window** ribbon tab.



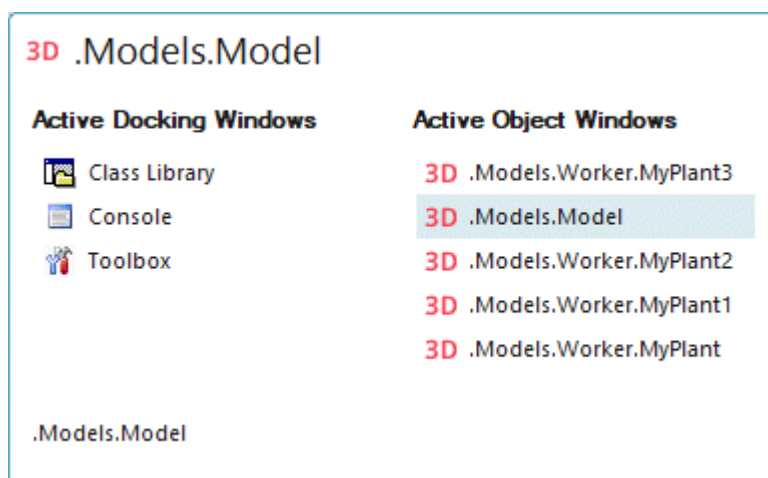
- To show or hide the *Favorites* window, click  on the **Window** ribbon tab.

- To show or hide the *Console*, click  on the **Window** ribbon tab.

- To show or hide the *Toolbox*, click  on the **Window** ribbon tab.

- To customize the **Quick Access Toolbar** and the **Ribbon**, click the down arrow on the **Quick Access Toolbar** and select **More Commands**.

- To **change between docking windows and object windows**, press **Ctrl+Tab**.



The **Active Docking Windows** are listed on the left and the **Active Object Windows** on the right.

- The **left arrow** and the **right arrow** change between the columns **Active Docking Windows** and **Active Object Windows**.
- The **Tab key** or the **up/down arrows** change between the windows within a column. Releasing **Ctrl** activates the selected window.

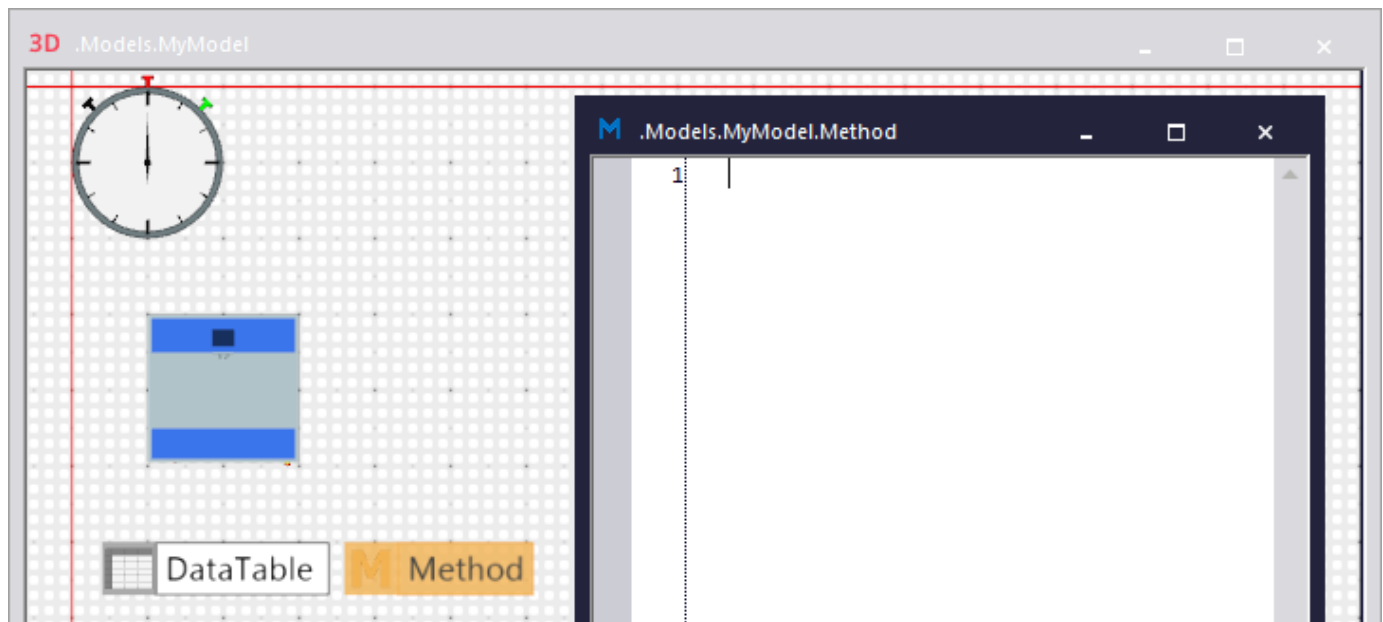
Back to [Work with Window Types](#)

Object Windows

Object windows are windows which you can maximize and minimize and arrange within the *Plant Simulation* program window.

Plant Simulation object windows are the windows of the objects **Frame** , **Method M**, **Method Debugger**, **DataQueue** , **DataStack** , **DataList** , **DataTable** , and the **Icon Editor**.

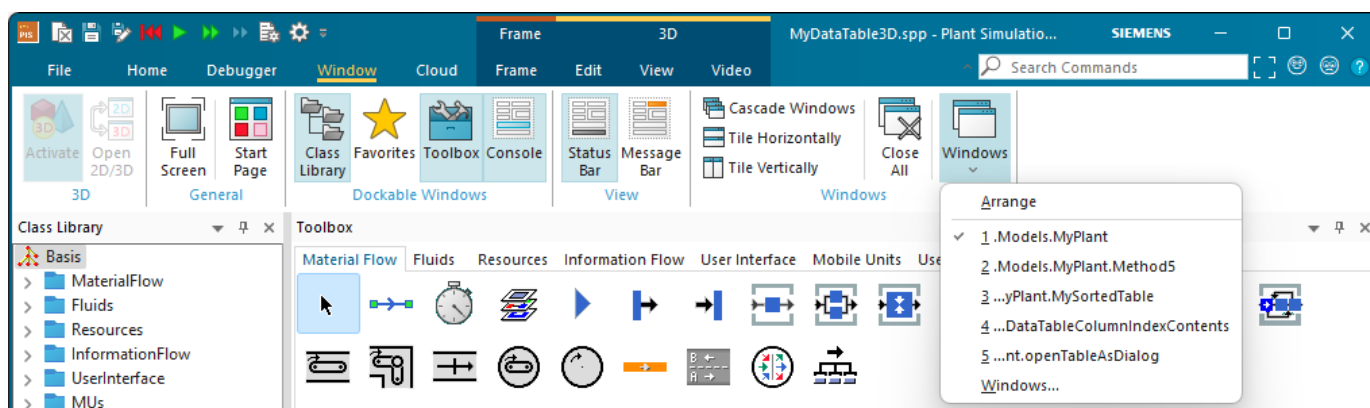
Object windows always open in the background. *Object windows* have a blue border in the figure below [Work with Window Types](#).



To bring an open object window to the foreground, click the icon of the window in the window tab bar.

To arrange the windows, i.e., to cascade, to tile them horizontally, or to tile them vertically, you can select the respective command on the **Window Ribbon Tab**.

To bring an open object window to the front, select the name of the window on the **Window** ribbon tab. You can also arrange these windows with one of these commands.



Note

You can open the *DataList*, the *DataQueue*, the *DataStack*, and the *DataTable* in the foreground as a window with the method `openDialogBox`, compare [Opening a List as a Dialog in the Foreground](#).

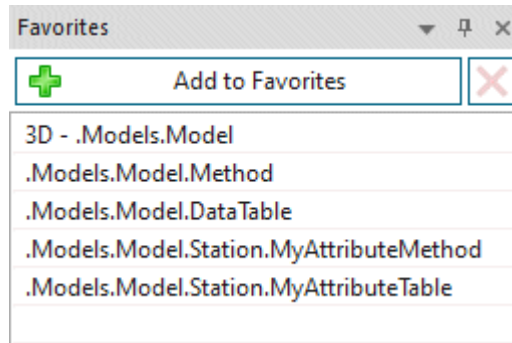
You can close all **object windows** with the function `closeAllWindows`.

You can minimize and maximize an *object window* and move it around within the program window. When you maximize an *object window*, *Plant Simulation* adds the button combination **Minimize**, **Maximize**, **Close**  that is otherwise located in the title bar of the window, above the *Ribbon Bar*.

You can add links to the following object windows to the **Favorites**:

- **Frame** windows .
- **Method** windows .
- **DataList** windows .
- **DataQueue** windows .
- **DataStack** window .
- **DataTable** window .
- Windows of user-defined attributes of data type *method*, *table*, *list*, *stack*, and *queue*.

You can then quickly open the respective window by double-clicking its name in the favorites list.

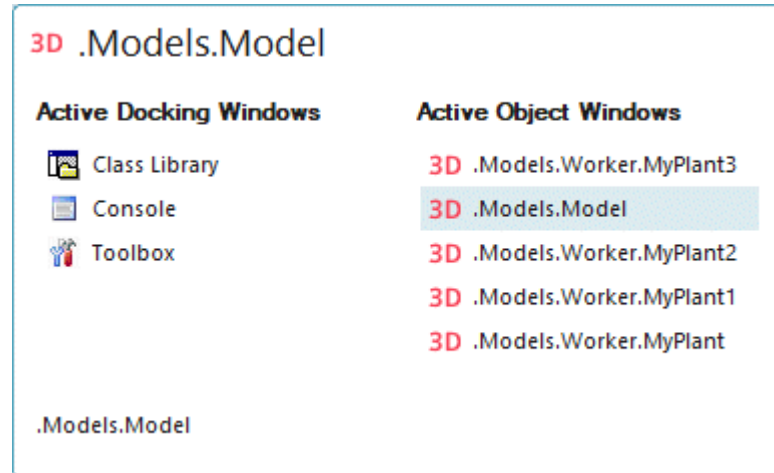


- To **change between docking windows and object windows**, press **Ctrl+Tab**.
 - The **left arrow** and the **right arrow** change between the columns **Active Docking Windows** and **Active Object Windows**.
 - The **Tab key** or the **up/down arrows** change between the windows within a column. Releasing **Ctrl** activates the selected window.

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Changing Between Docking and Object Windows

Press **Ctrl+Tab** to open the **Window Navigator**. It shows a preview of the docking windows and of the opened object windows. The windows navigator stays open until you release **Ctrl** again.



The **Active Docking Windows** are listed on the left and the **Active Object Windows** on the right.

- The **left arrow key** and the **right arrow key** change between the columns **Active Docking Windows** and **Active Object Windows**.
- The **Tab key** or the **up/down arrow keys** change between the windows within a column. Releasing the **Ctrl key** activates the selected window.

If the focus is located on a docking window, **Ctrl+Tab** changes between the docking windows.

If the focus is located on an object window, **Ctrl+Tab** changes between the object windows.

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Dialog Boxes

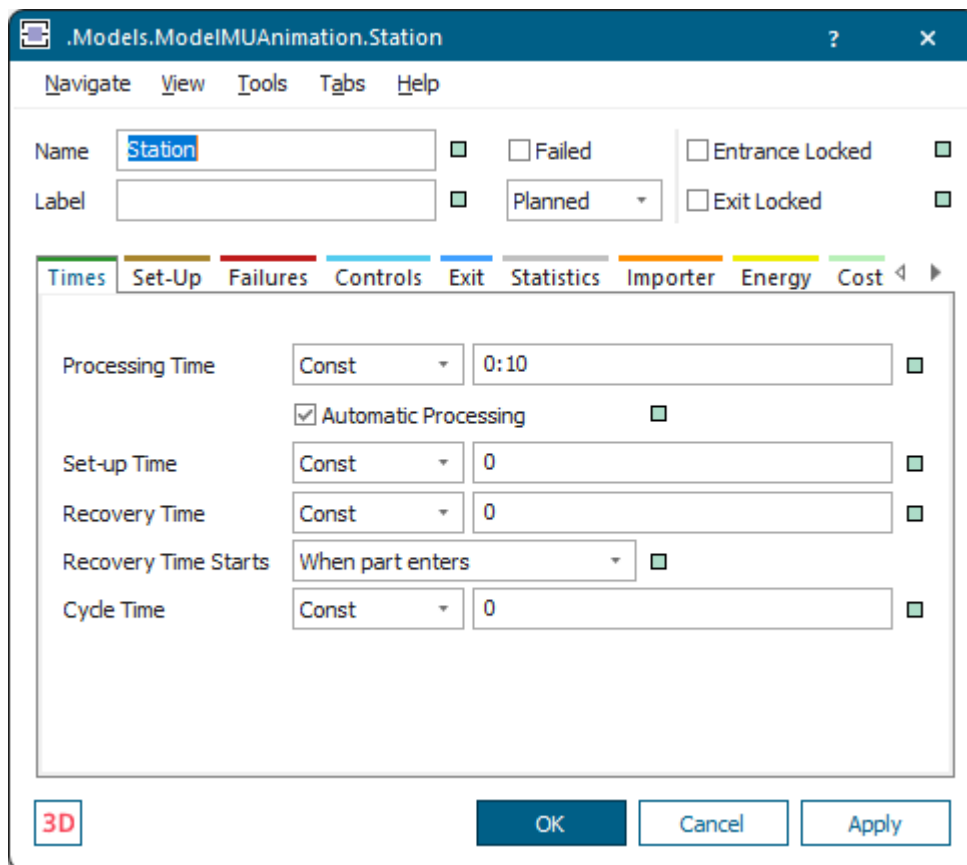
Plant Simulation dialog boxes are the *dialog boxes* of the *Plant Simulation material flow objects*, the *mobile objects*, the *fluid objects*, the *resource objects*, the *information flow objects*, and the *user interface objects*.

Dialog boxes have a green border in the figure below [Work with Window Types](#).

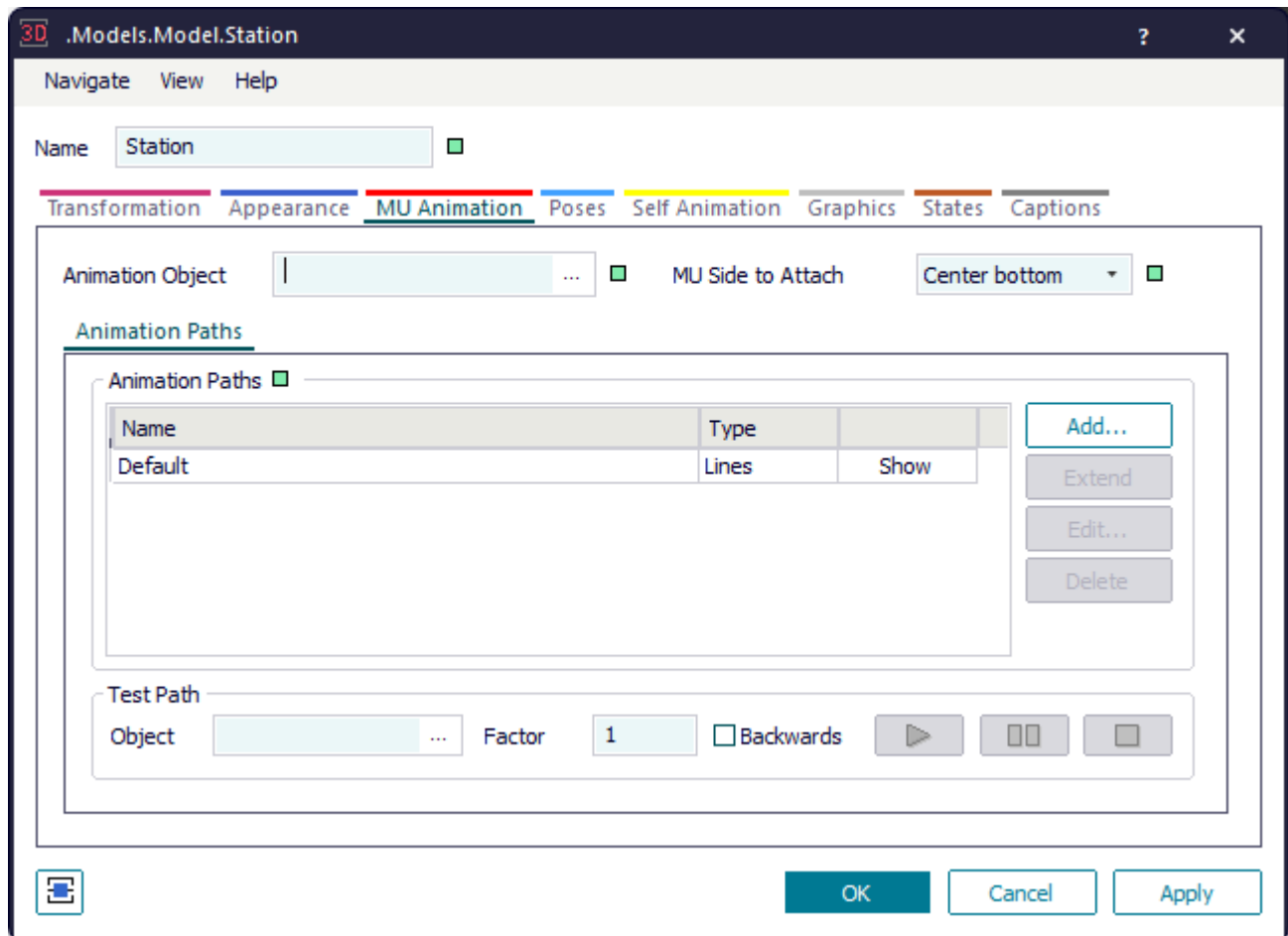
A *dialog box* always opens in the front as it expects that you select or enter something. You cannot minimize or maximize it.

You can drag a *dialog box* outside the program window and move it around on the entire screen.

In the *dialog boxes* of the objects you can edit their **simulation properties** on the respective tabs.



You can open the dialog box [Edit 3D Properties](#) by clicking the button  on the left bottom corner of the dialog box to edit the **3D animation properties**.



To close all open document windows at once, click **Close All Windows**  on the **Window** ribbon tab.

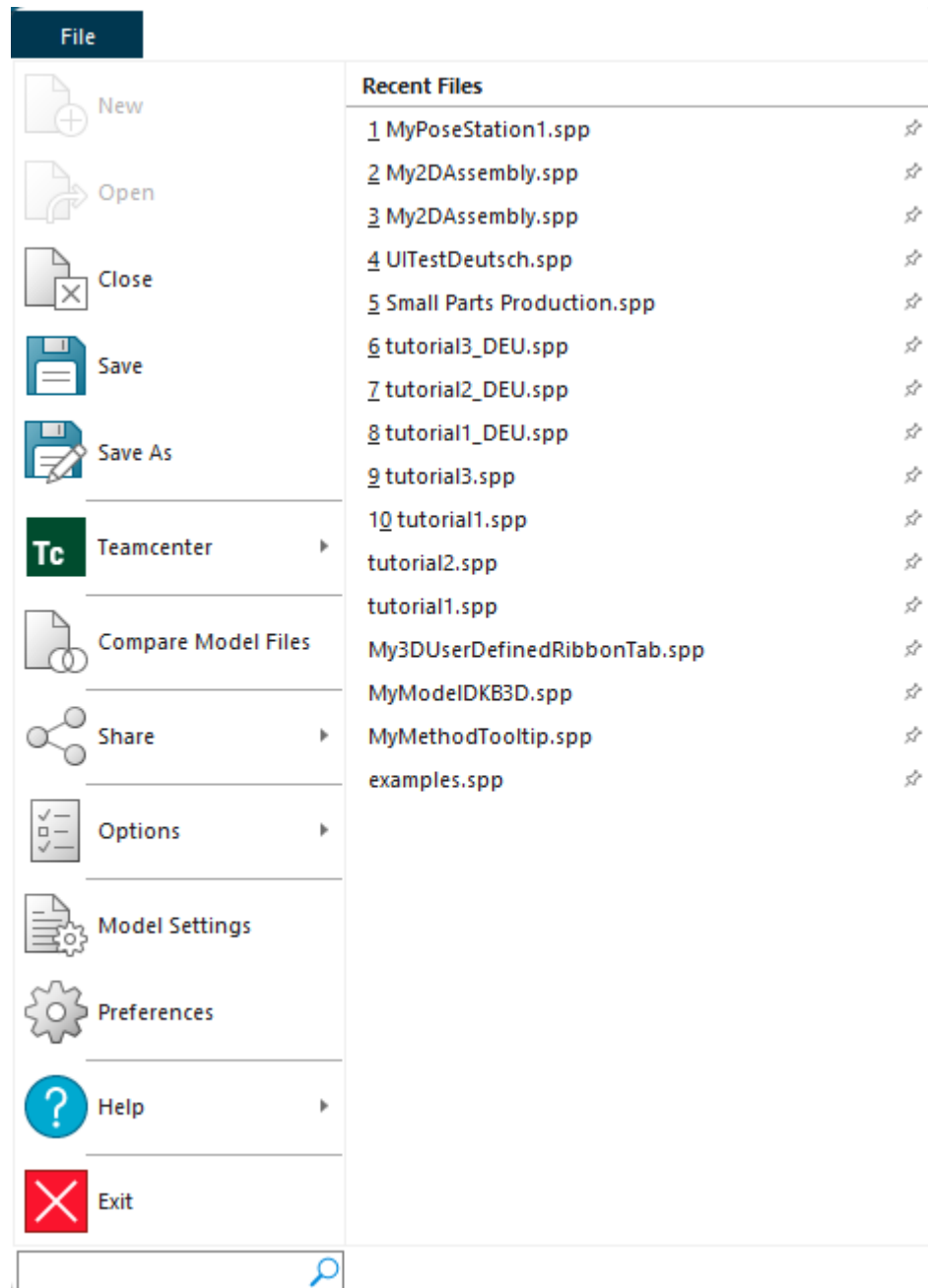
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Selecting Settings in Plant Simulation

Select Settings in Plant Simulation

Before you start working with *Plant Simulation*, you might want to change some of the default settings that affect your simulation model.

- To select settings, which apply to the **active model**, select **File > Model Settings**. *Plant Simulation* saves these settings in the model file.



- To select settings, which apply to **new models**, select **File > Preferences**. On the tabs **General [preferences]**, **3D [preferences]**, and **Editor [preferences]** you can select general settings, which are independent of the simulation model. On the tabs **Simulation [preferences]**, **User Interface**, and **Units [preferences]** you can select model-specific settings, which apply to new models.

You can:

- Select General Options**
- Select Options for Units and for Displaying the Time**

Note

You can add commands, that you often use, to the [Quick Access Toolbar](#)



Compare [Preferences](#)

Compare [Select View Options in the Frame](#)

Back to [Working with Plant Simulation, Basics](#)

Select General Options

Select the language *Plant Simulation* uses when it creates a new simulation model. This language determines the names of folders and objects, such as *MaterialFlow* or *Materialfluss*, *Station* or *Einzelstation*, etc.

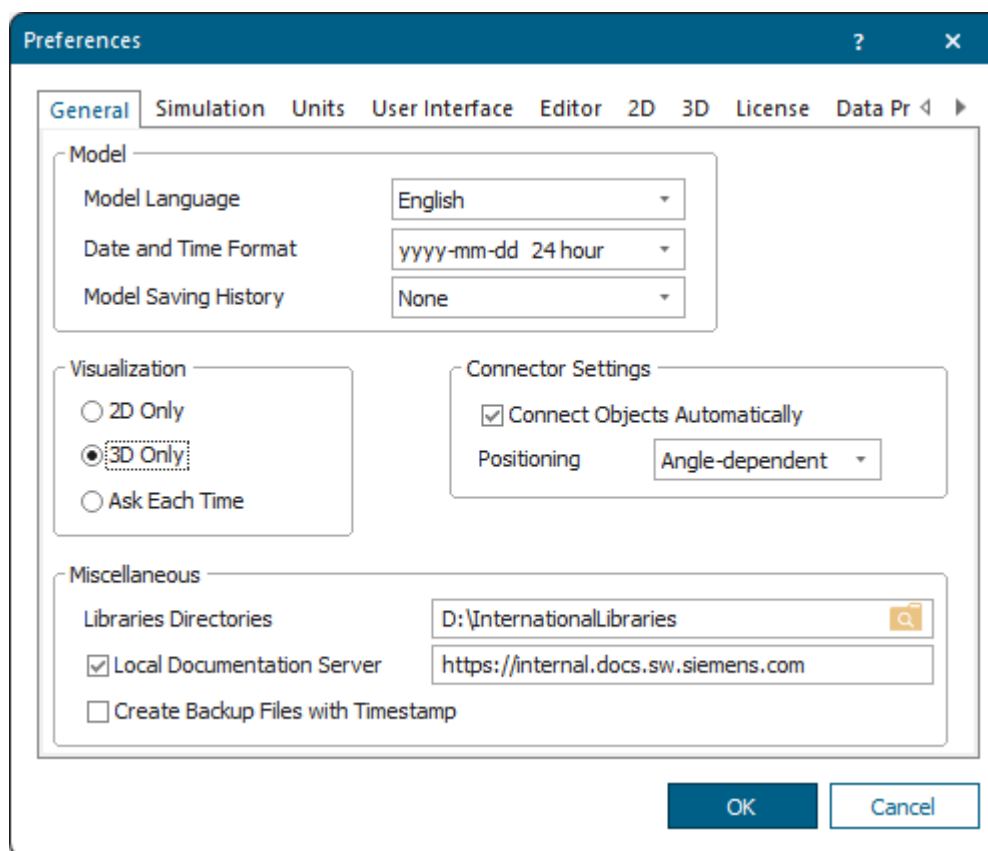
Note

The language of the operating system of your computer determines the language of the *Plant Simulation* user interface.

The language determines:

- The name of the items of the drop-down lists, such as the names of the distributions (**Uniform** or **Gleich**).
- The return value of the attributes that correspond to the entries of the drop-down lists, such as *Station.ProcTime.Type*.
- The contents of the window **Show Attributes and Methods**.
- The names which the feature **Auto Complete** within the *Method Editor* suggests.

Select if you want to use the **24-hour clock**, the **12-hour clock (AM/PM)**, or the **DMY** (day-month-year) notation date and time format on the **General** ribbon tab.



Date and time format	Looks like this
yyyy/mm/dd 12 hour	Variable=2024/01/01 12:00:00.0000 AM
yyyy-mm-dd 24 hour	Variable=2024-01-01 00:00:00.0000
dd.mm.yyyy 24 hour	Variable=01.01.2024 00:00:00.0000

Select which kind of comment *Plant Simulation* adds to your model file each time you save it:

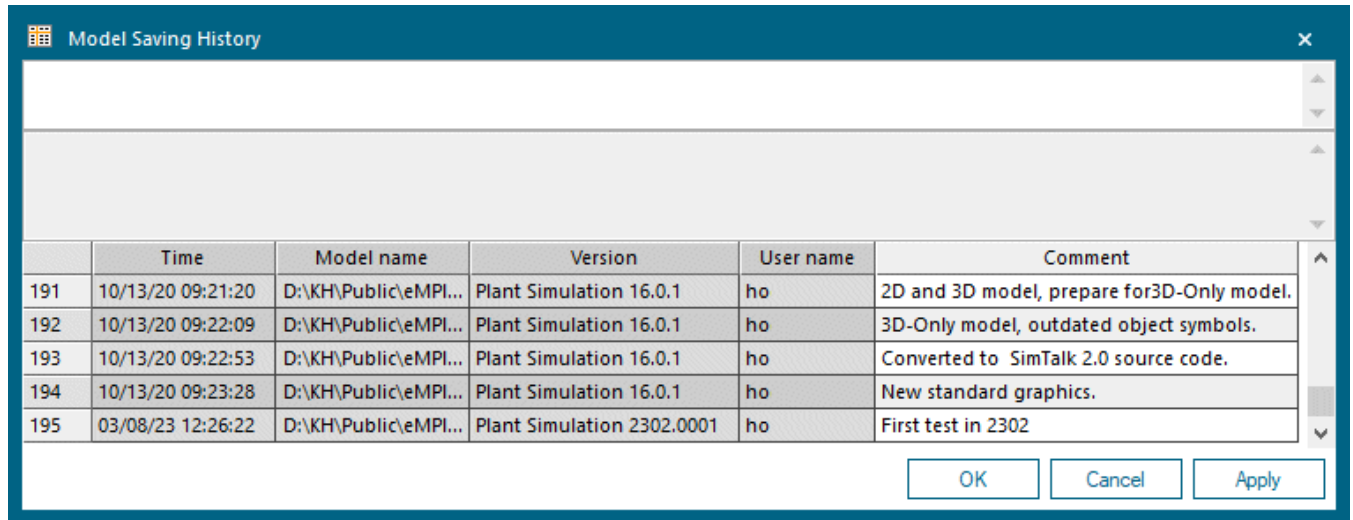
- **Without comment** adds no comment to the model (.spp) file.
- **With comment** opens a comment window each time when you save the model. Enter your comment here.
- When you select **Without comment** or **With comment** *Plant Simulation* adds a row to the dialog **Model Saving History** each time you save the model.

When the model crashes, *Plant Simulation* adds a description of the problem to the model history window instead of a comment. This helps our software engineers detect what caused the crash.

- **None** does not save the history of how you saved your model.

To view the comment you entered:

Select **File > Options > Show Model History** in the program window.



The image shows a 'Model Saving History' dialog box with a table of saving records. The table has six columns: an index, Time, Model name, Version, User name, and Comment. It contains five rows of data. At the bottom right of the dialog are three buttons: OK, Cancel, and Apply.

	Time	Model name	Version	User name	Comment
191	10/13/20 09:21:20	D:\KH\Public\eMPI...	Plant Simulation 16.0.1	ho	2D and 3D model, prepare for 3D-Only model.
192	10/13/20 09:22:09	D:\KH\Public\eMPI...	Plant Simulation 16.0.1	ho	3D-Only model, outdated object symbols.
193	10/13/20 09:22:53	D:\KH\Public\eMPI...	Plant Simulation 16.0.1	ho	Converted to SimTalk 2.0 source code.
194	10/13/20 09:23:28	D:\KH\Public\eMPI...	Plant Simulation 16.0.1	ho	New standard graphics.
195	03/08/23 12:26:22	D:\KH\Public\eMPI...	Plant Simulation 2302.0001	ho	First test in 2302

Compare [Creating a German Model with English Identifiers](#)

Back to [Select Settings in Plant Simulation](#)

Select Options for Units and for Displaying the Time

When creating your simulation models, you may have to choose settings for the units you are going to use and the settings for displaying the time.

As daylight saving time is widely used internationally, you will also use it in your simulation model.

The screenshot shows the 'Units' tab in a simulation software interface. The 'Units' section includes dropdown menus for Mass (kg), Currency (€), Speed (m/s (mps)), Acceleration (m/s² (mps²)), Length (m), and a field for 1 LU = 1 m. The 'Daylight Saving Time' section has a checkbox and a table for Start and End times. The 'Time Scale' section has a Time Scale field (1.000000) and a Transfer If field (24 : 60 : 60).

	Month	Week	Day	Hour
Start	3	-1	0	2
End	10	-1	0	3

Note

To select units and time settings, which only apply to the active model, select **File > Model Settings > Units**.

To select units and time settings, which apply to all new models, select **File > Preferences > Units**.

- Select the check box **Daylight Saving Time**.
- If the plant you are modeling is located in the **European Union**, you do not have to change the default settings. Here daylight saving time starts at 02:00 o'clock in the morning Greenwich Mean Time (GMT) on the last Sunday in March. It ends at 03:00 o'clock in the morning on the last Sunday in October. In the EU, all time zones change at the same moment.
- If the plant you are modeling is located in the **United States of America**, you have to change the settings. For most of the US daylight saving time starts at 02:00 o'clock in the morning on the second Sunday of March. It ends at 02:00 o'clock in the morning on the first Sunday of November.

☒ Daylight Saving Time

	Month	Week	Day	Hour
Start	4	-1	0	2
End	10	-1	0	2

If your specific modeling needs require it, you can also change the display of the **Time Scale**. You can enter a number between 0 and 86400. You can enter an integer greater than 1 into the text boxes next to **Transfer If**.

You might, for example, divide a minute into 100 units, instead of in 60 seconds, or you might want to read 1:50 as 1.5 hours instead of 1 minute and 50 seconds.

Note

You can only change the display of the time scale, not the time scale itself. You can only change these settings if no simulation model is open!

A time statement consists of four numbers, separated by colons, normally in the format **days:hours:minutes:seconds**. *Plant Simulation* stores the value itself in seconds and converts it at output time for the **display**.

Plant Simulation converts times according to the values you enter into the text boxes **Time Scale** and **Transfer If**.

The following examples illustrate what to enter to achieve different displays of the time:

- If you want to use the standard time format, 24 hours to the day, 60 minutes to the hour, 60 seconds to the minute, use the default setting **Time Scale 1/1.0 Transfer If 24:60:60**.
- If you want to simulate a longer period of time and want to display the time in the format **years:months:days:hours**, enter **Time Scale 1/3600 Transfer If 12:30:24**, as an hour has 3600 seconds, a year has 12 months, a month has 30 days, and a day has 24 hours.
- If you want to divide a minute into 100 subunits instead of 60 seconds, enter **Time Scale 1/0.6 Transfer If 24:60:100**.

Compare [Select View Options in the Frame](#)

Compare [Units \[preferences\]](#)

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Changing the Settings of the Objects

Change the Settings of the Objects

You can change the settings of the objects, i.e., the values of attributes, in a number of ways.

You can:

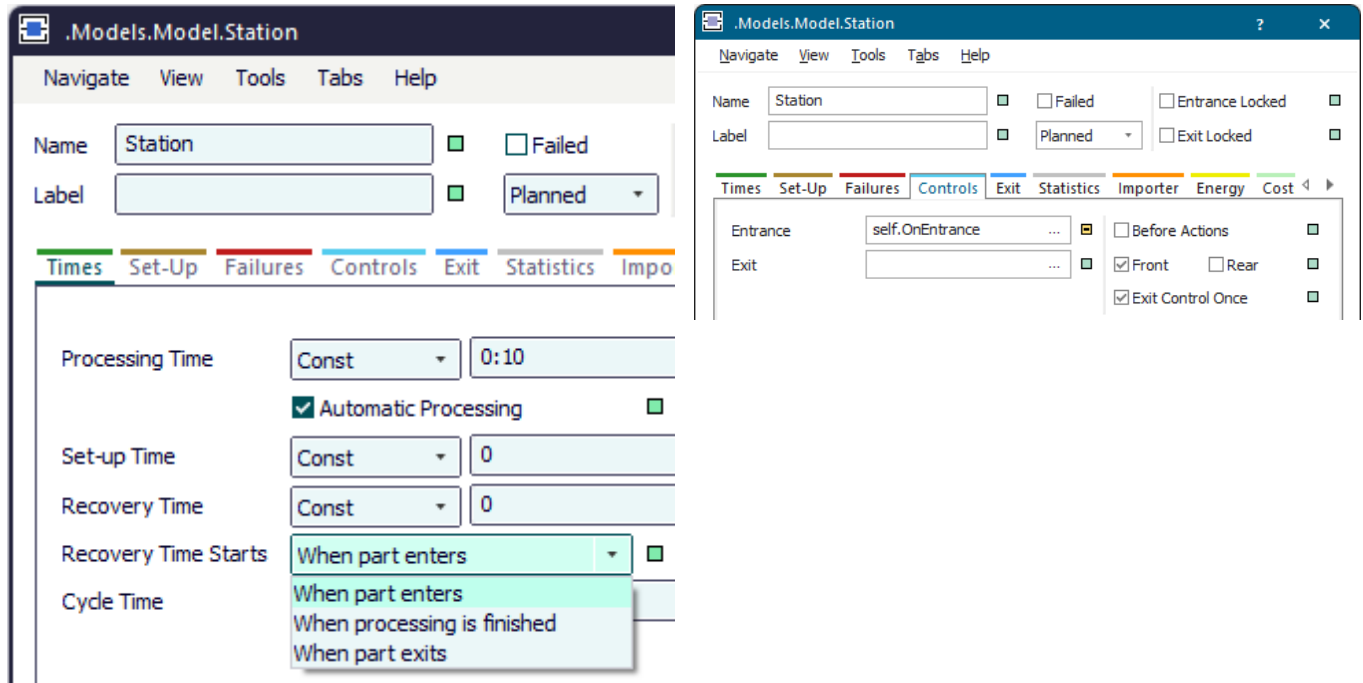
- [Change Values in the Dialog of the Objects](#)
- [Change Values by Assigning a Value in SimTalk](#)
- [Change Values in the Window Show Attributes and Methods](#)
- [Change Values with the AttributeExplorer](#)

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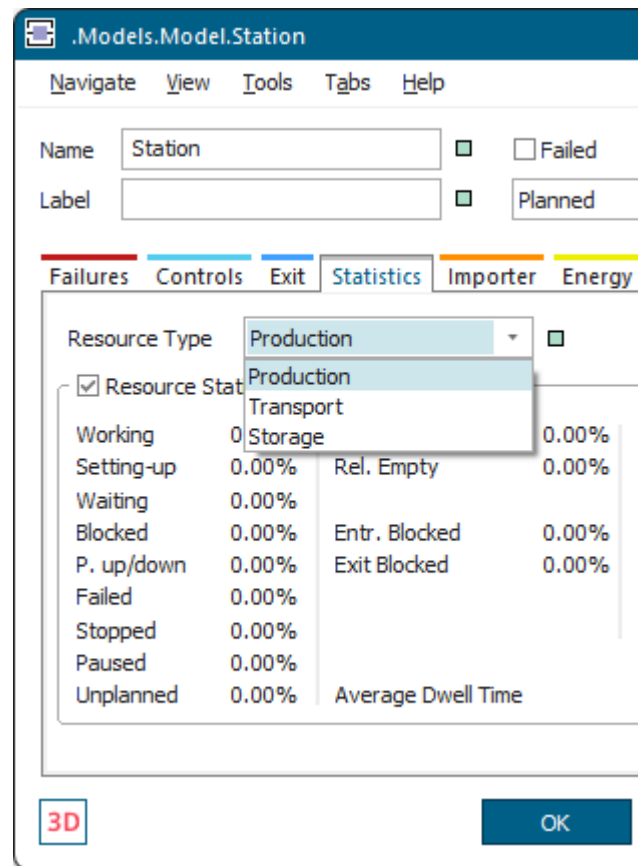
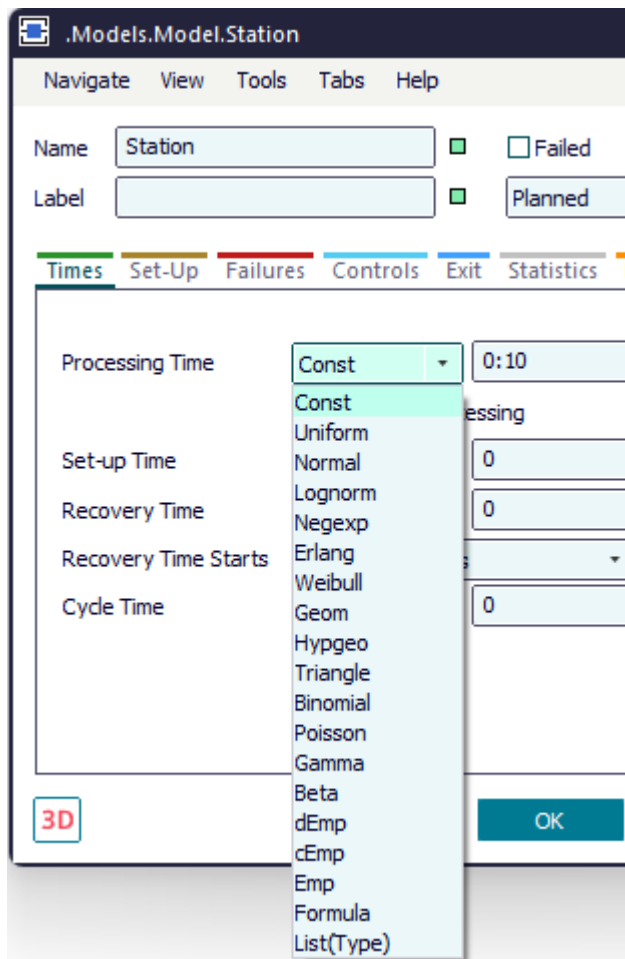
Change Values in the Dialog of the Objects

As a rule you are going to select the settings of the individual objects in your simulation model in the dialogs of the respective objects.

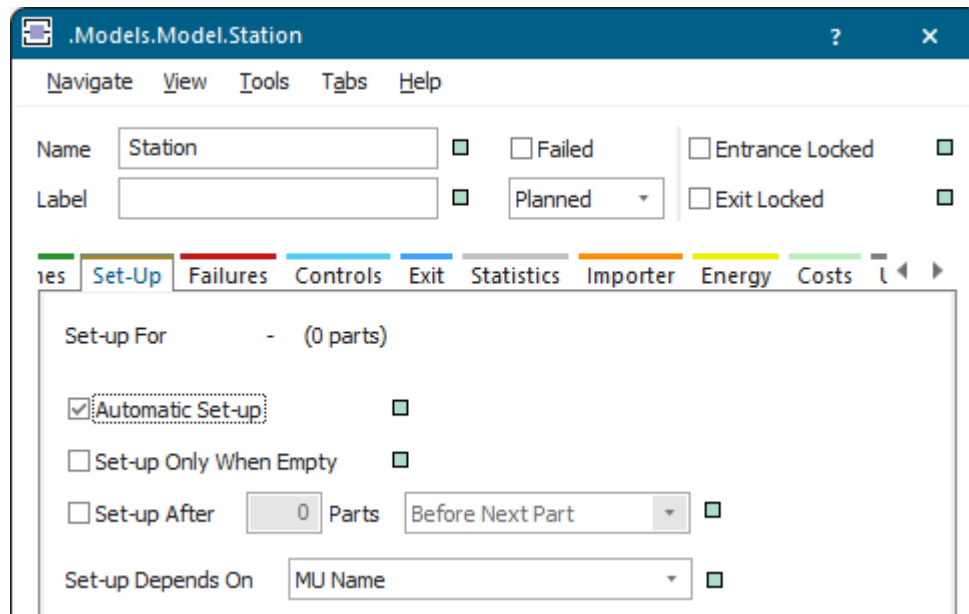
To do so, you either **enter a value into a text box**.



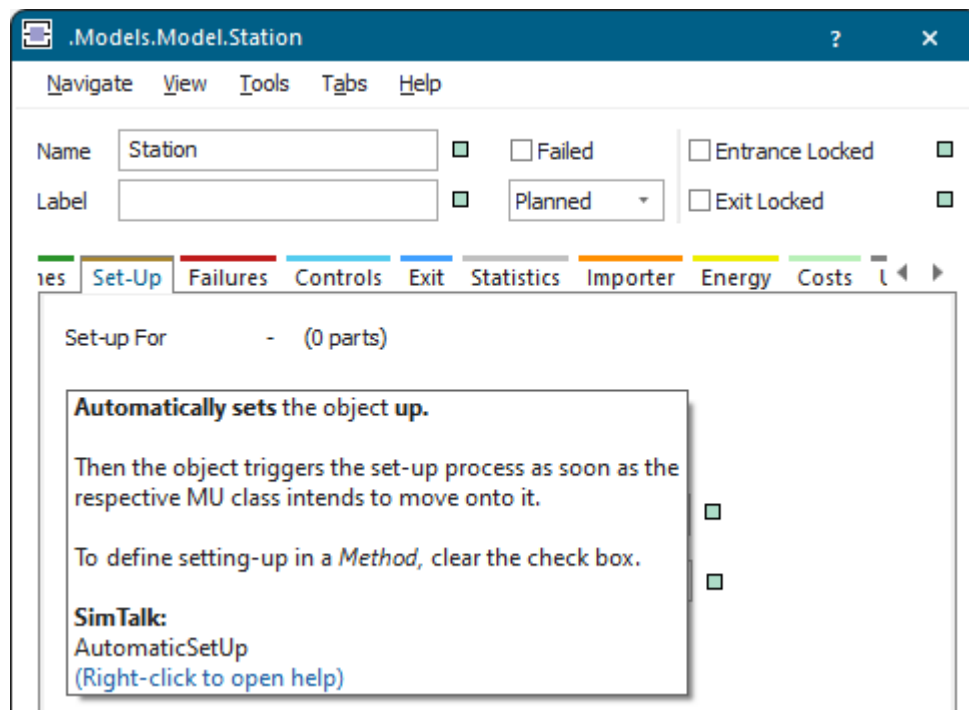
Or you **select a value from a drop-down list**.



Or you **check or clear** a **check box** to activate or to deactivate a setting. Instead of clicking the **check box** itself, you can also click its **label**.



All of these settings correspond to *attributes* which you can also access with the programming language *SimTalk*. The *What's This* help for the respective dialog item shows the name of the *attribute* in the bottom row below **SimTalk**.

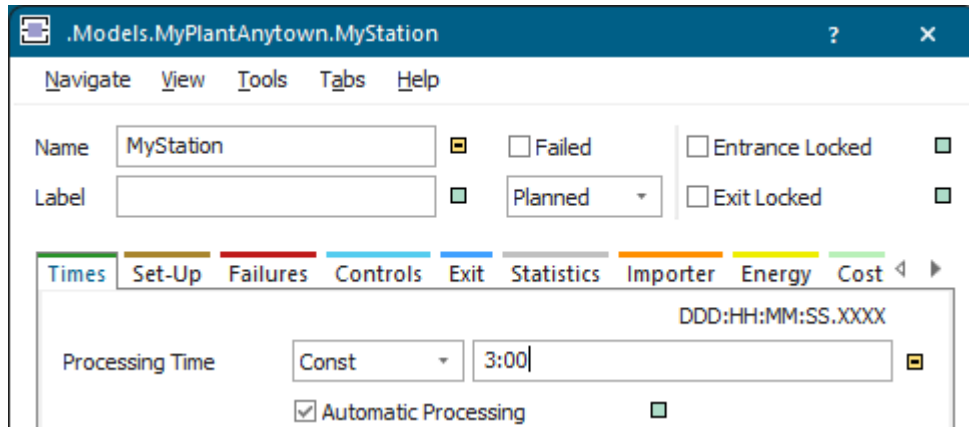


Back to [Change the Settings of the Objects](#)

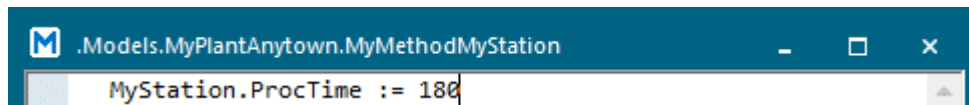
Change Values by Assigning a Value in SimTalk

You can change the settings, which you can select in or type into the dialogs of the objects, by assigning a value to an object attribute in *SimTalk*.

To set a constant **Processing Time** of **three minutes**, you can type the setting below into the dialog.



Or you can type this instruction into a *Method*:



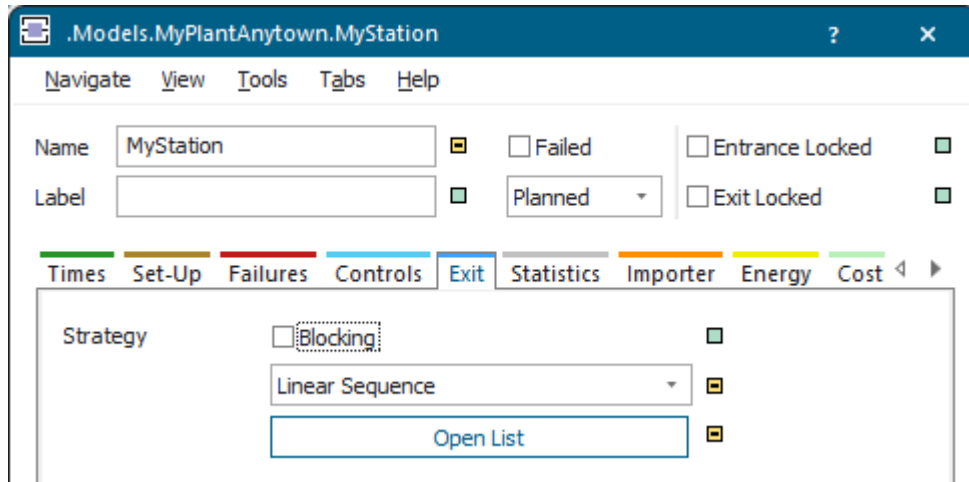
Paths in SimTalk Instructions

In the example we inserted the material flow object *MyStation* into the *Frame* called *Models*. This is why the title bar of the station *MyStation* shows the absolute path **.Models.MyPlantAnytown.MyStation**.

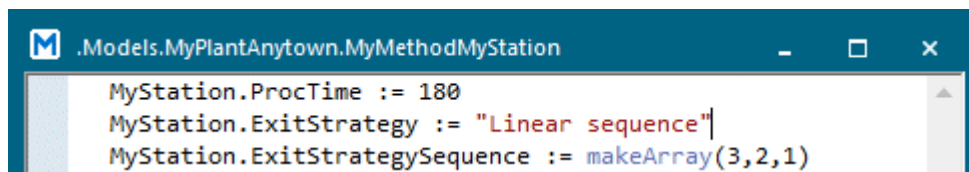
As we inserted the *Method* named *MyMethodMyStation* into the same *Frame* as the station *MyStation*, which the title bar of the *Method* also shows, we can type in the instruction `MyStation.ProcTime := 180` or `MyStation.ProcTime := 3:00` and do not have to type the entire path, i.e., `.Models.MyPlantAnytown.MyStation.ProcTime := 180`.

Within the instruction `MyStation.ProcTime := 180` the term *MyStation* designates the object for which the instruction applies. This is followed by a separator, namely a period `.`, followed by the name of the attribute which we want to change, *ProcTime* in our case. The value after the assignment operator `:=`, without a blank space between the colon and the equal sign, is assigned to the value of the attribute while the is being executed, 180, i.e., three minutes in our case.

To set the **Exit Strategy > Linear sequence**, you can select it in the drop-down list and then type the numbers of the successors of the *Station* into the **List** that clicking the button **Open List** opens.



Or you can type these instructions into a *Method*:



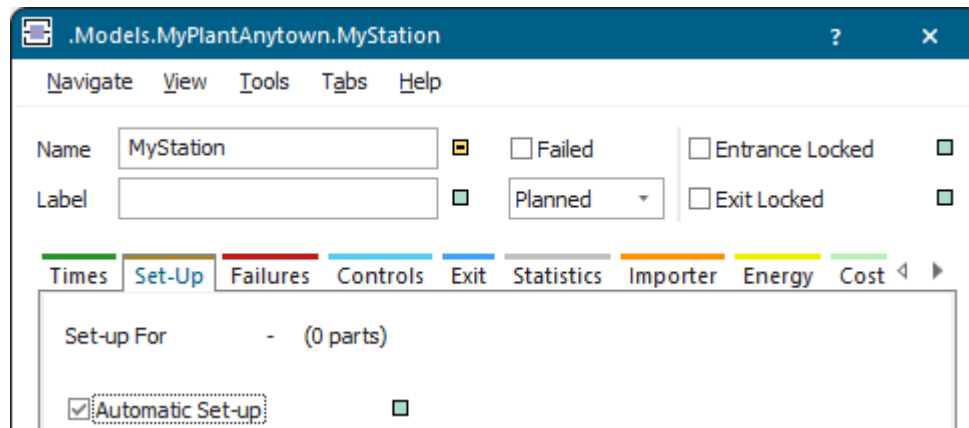
Note

When you type a *string* into an assignment, *Plant Simulation* does not differentiate between sentence case and lower case.

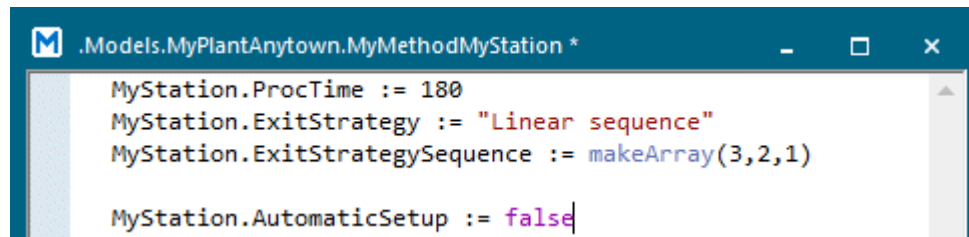
When you query the value, it shows it in sentence case as the respective drop-down list shows it. In the example above you could type "Linear sequence", "linear sequence", "LINEAR SEQUENCE", etc.

When you query the value with `print`, *Plant Simulation* shows **Linear sequence**.

To manually set the *Station* up, clear the check box **Automatic**.



Or you can type this instruction into a *Method*:



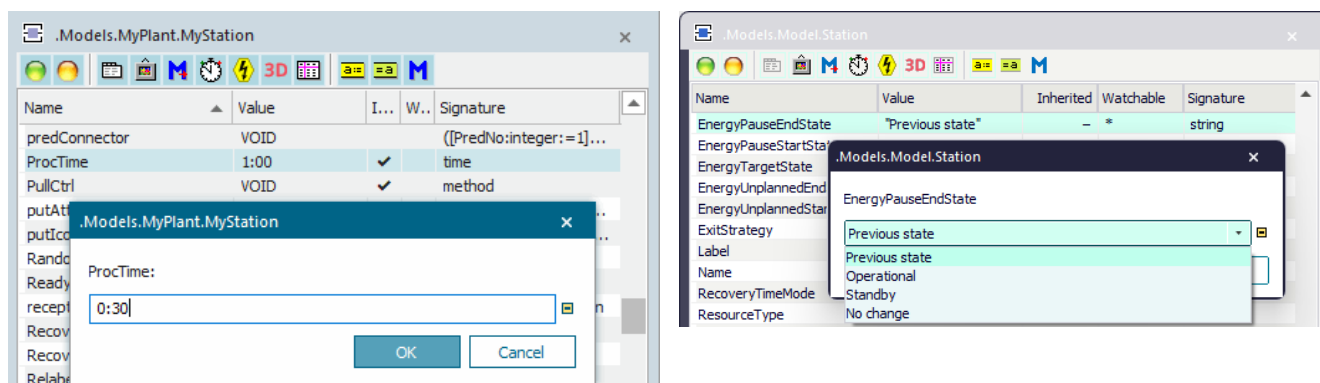
Back to [Change the Settings of the Objects](#)

Change Values in the Window Show Attributes and Methods

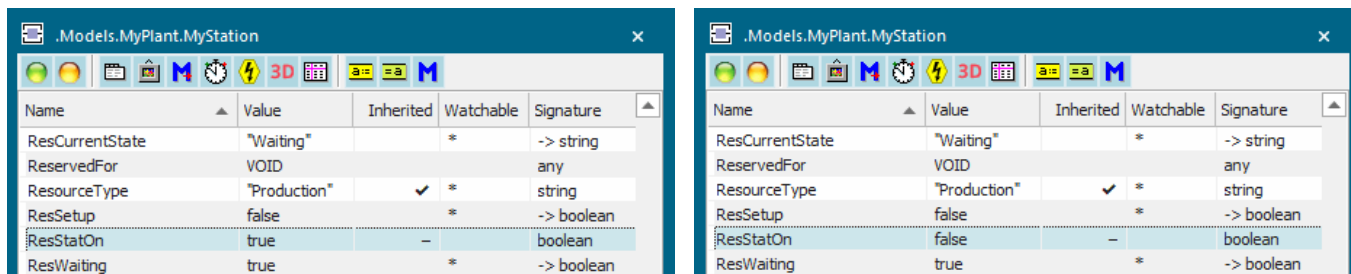
You can access the values of the settings of the objects in the window **Show Attributes and Methods**.

To change attributes in the window **Show Attributes and Methods** of the respective object, proceed as follows:

- To open the dialog, click the object and press the **F8** key.
- Then, double-click the respective attribute.
- Enter the new value into the dialog that opens or select a value in the drop-down list.



To switch the value of an attribute of data type *boolean* from **true** to **false** and vice versa, double-click the row of the attribute in the window **Show Attributes and Methods**.



Compare Simultaneously Show and Change Attributes of Several Objects

Compare Working with the Window Show Attributes and Methods

Back to Change the Settings of the Objects

Change Values with the AttributeExplorer

Instead of opening the dialog of each and every material flow object in your model and entering values for a single attribute or for a number of attributes into the text boxes, you can define which attributes of which objects the *AttributeExplorer* gets.

Using the **AttributeExplorer**  you can manage the attributes defining the settings of the individual stations in your simulation model at a single location. You can then enter different values for the capacities, the times, etc. *Plant Simulation* writes these values back to the dialogs of the objects and uses them in your simulation model. You can also export this settings table as a tab-delimited text file and import this file into the *AttributeExplorer* of another simulation model, thus using identical settings in several of your simulation models.

You can also use the *AttributeExplorer* to find objects of the type you define, to find attributes, for example the position of a number of objects. You can then align them in the *Frame*, etc.

Compare Set Parameters with the AttributeExplorer

Compare Specify the Attributes You Want to View or Change

Back to Change the Settings of the Objects

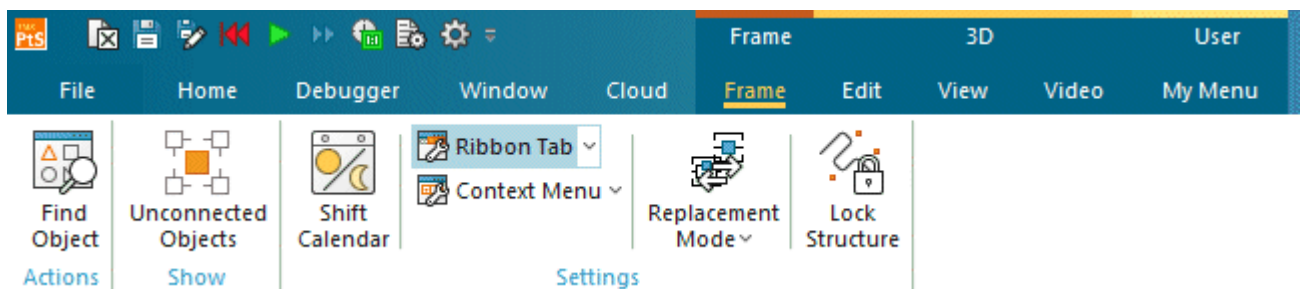
Finding Objects and Text in Your Simulation Model

Find Objects and Text in Your Simulation Model

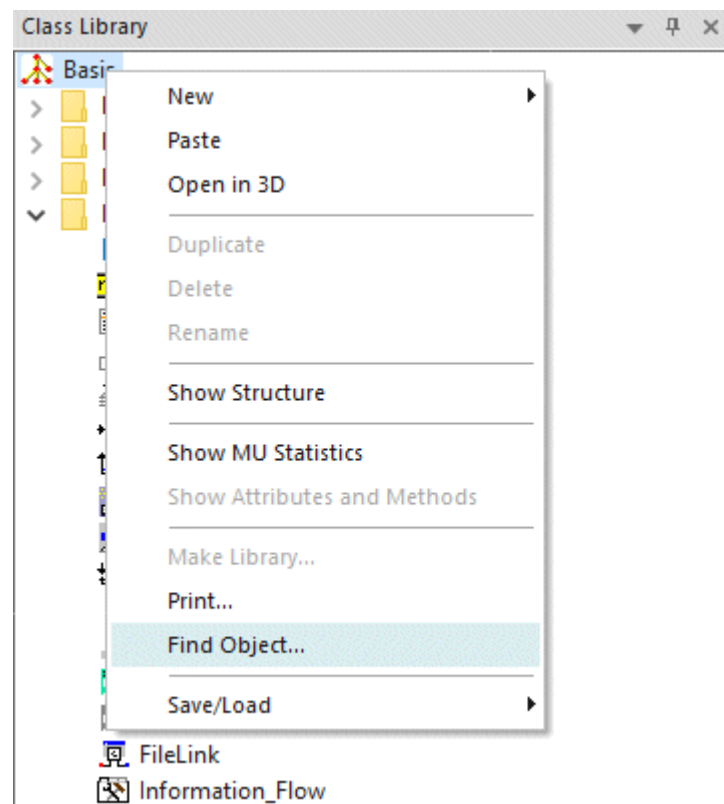
You can find the **name** of an object, a **condition**, an expression contained in an **attribute**, part of the **source code** of a *Method*, or an expression contained in a **table** in your simulation model.

Proceed as follows:

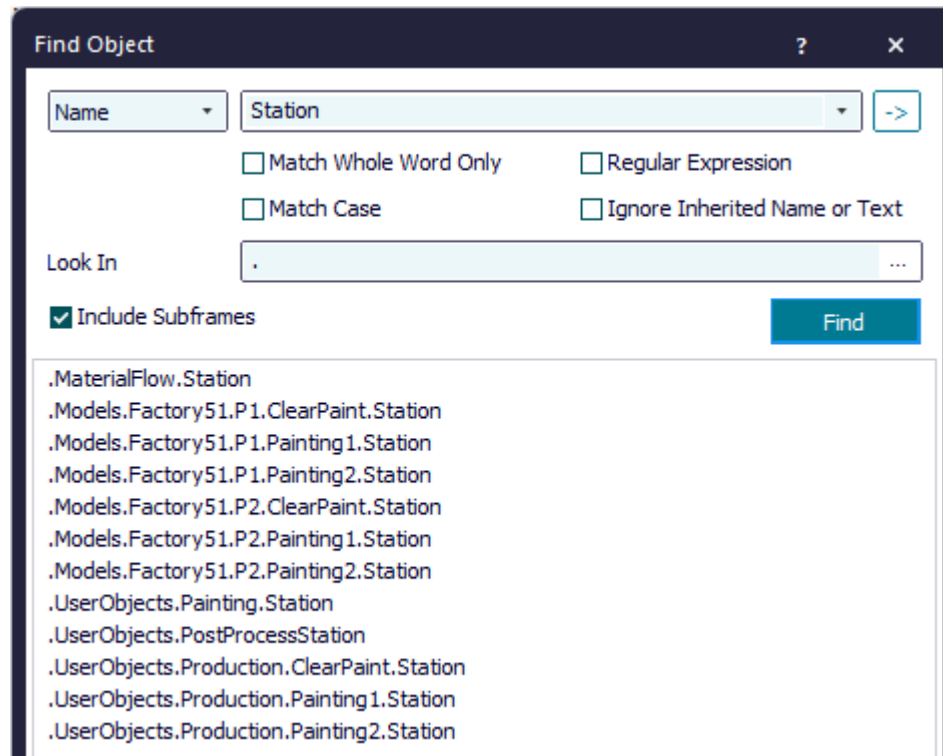
1. Click the **Frame** ribbon tab and click **Find Object**.



Or click the right mouse button anywhere in the *Class Library* and select **Find Object** on the context menu.



2. Select what you want to find from the left drop-down list.



Compare [Find Objects and Attributes](#) using the AttributeExplorer

Back to [Working with Plant Simulation, Basics](#)

Video on YouTube

<https://youtu.be/FNpiAaUK8iM?si=B1tgMoMpdC03wCmj&t=437>

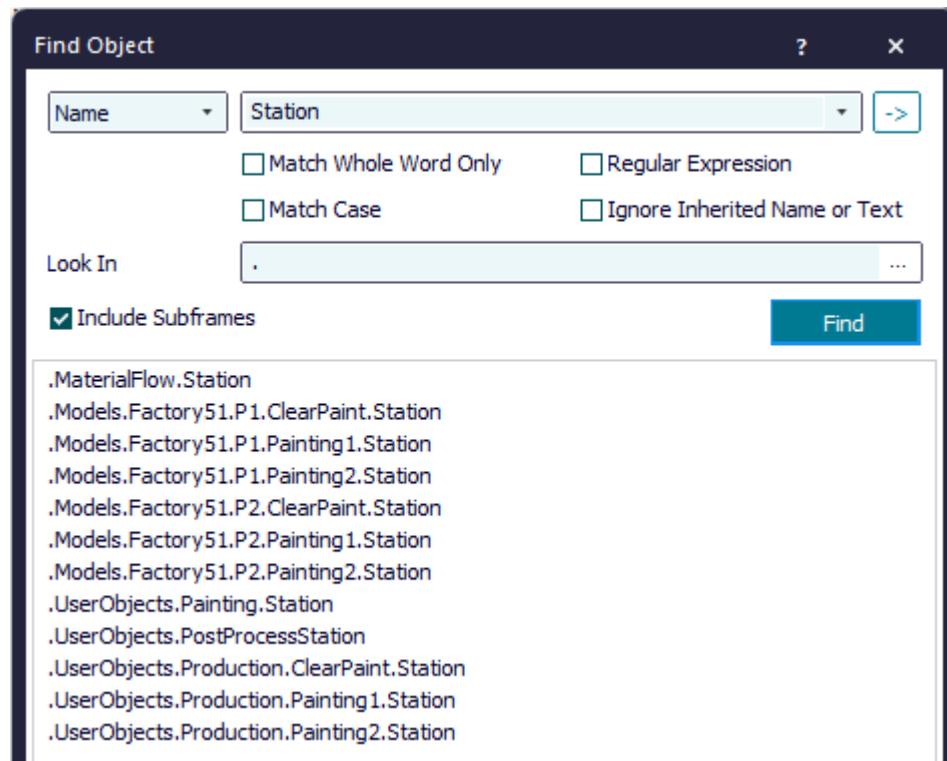
Find the Name of an Object

You can find the name of an object you inserted into your simulation model.

Proceed as follows:

1. Select **Name** from the left drop-down list.
2. Type in the name of the object you want to find into the right drop-down combo box. We entered **Station**.
3. If you do not select the check box **Match Whole Word Only**, *Plant Simulation* finds all words that contain the word you entered.

In our example it finds all word containing the word *Station*, such as *Station*, *Station1*, *Station11*, *MyStation*, etc.



4. Type the name of the folder or the *Frame* into the text box **Look In**, where *Plant Simulation* is going to start searching for the expression you entered into the text box.

Instead, you can also click **...** and select a folder or a *Frame* in the dialog **Select Object**.

To search on the highest level of the structure, accept the default setting that *Plant Simulation* enters, namely the period (.).

5. Select **Include Subframes** to also search all *Frames* located in the *Frame* in which the item you are searching for is located.
6. Click **Find**, to start searching.

7. Double-click the name of the object in the results field, to open its dialog. You can then change any of the settings.
8. If you select **Match Whole Word Only**, *Plant Simulation* only finds entire words that exactly match the word you entered. In our example it only finds the word *Station*.
9. If you select **Match Whole Word Only**, and also select **Match Case**, *Plant Simulation* only finds words that exactly match the word you entered, including upper and lower casing. In our example it only finds the word *Station*.
10. If you select **Regular Expression**, *Plant Simulation* also finds the regular expression you select. Click  and select any of the items displayed in the window **Regular Expression**. *Plant Simulation* enters the special characters designating that expression into drop-down combo box to the right.

Select	To enter and to find
Any single character	.
Beginning of line	^
End of line	\$
Beginning of word	\<
End of word	\>
Group	\()
Zero or one matches	?
Zero or more matches	*
One or more matches	+
Any one character in the set	[]
Any one character not in the set	[^]
Or	

Searching with regular expressions enables you to use wild cards in the string you are searching for.

- To find all strings that contain a sequence of the character a, followed by any character, and the character b, enter a . b into the text box.
- To find all objects whose name starts with an upper case S, enter ^[S].
- To find all objects whose name does not end with an e, enter [^e]\$.
- To find all objects whose name contains an upper case L, followed by any character, followed by proc, enter L . *proc into the text box.

11. Click **Find**.

Back to [Find Objects and Text in Your Simulation Model](#)

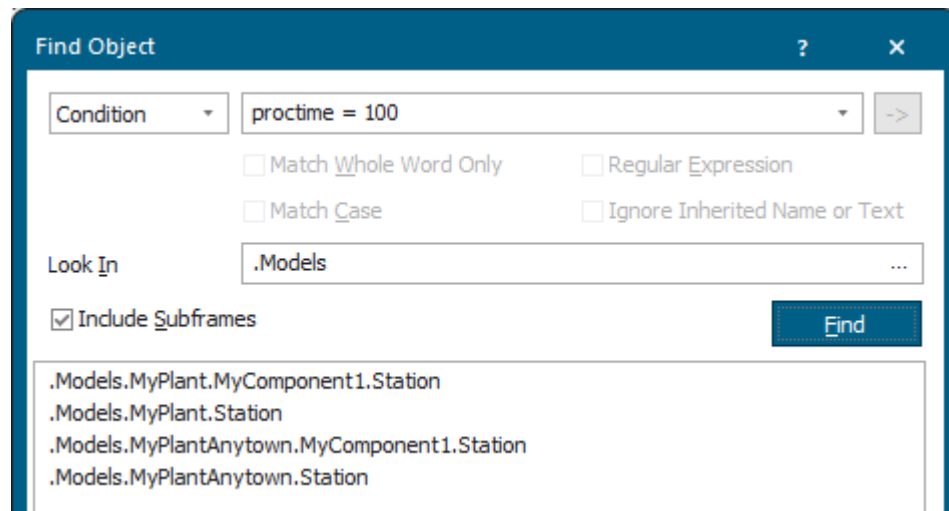
Find a Condition of an Object

You can find a condition of an object you inserted into your simulation model.

Proceed as follows:

1. Select **Condition** from the left drop-down list.
2. Enter any *SimTalk* expression whose *attributes* or *methods* are identical to the expression you enter in the drop-down combo box.

You might, for example, enter the name of an *attribute* or a *method* and the expression you are looking for, such as `proctime = 100`, or `OpenCtrl = void`, etc.



3. Repeat steps 4 to 7 described under [Find the Name of an Object](#)
4. Click **Find**.

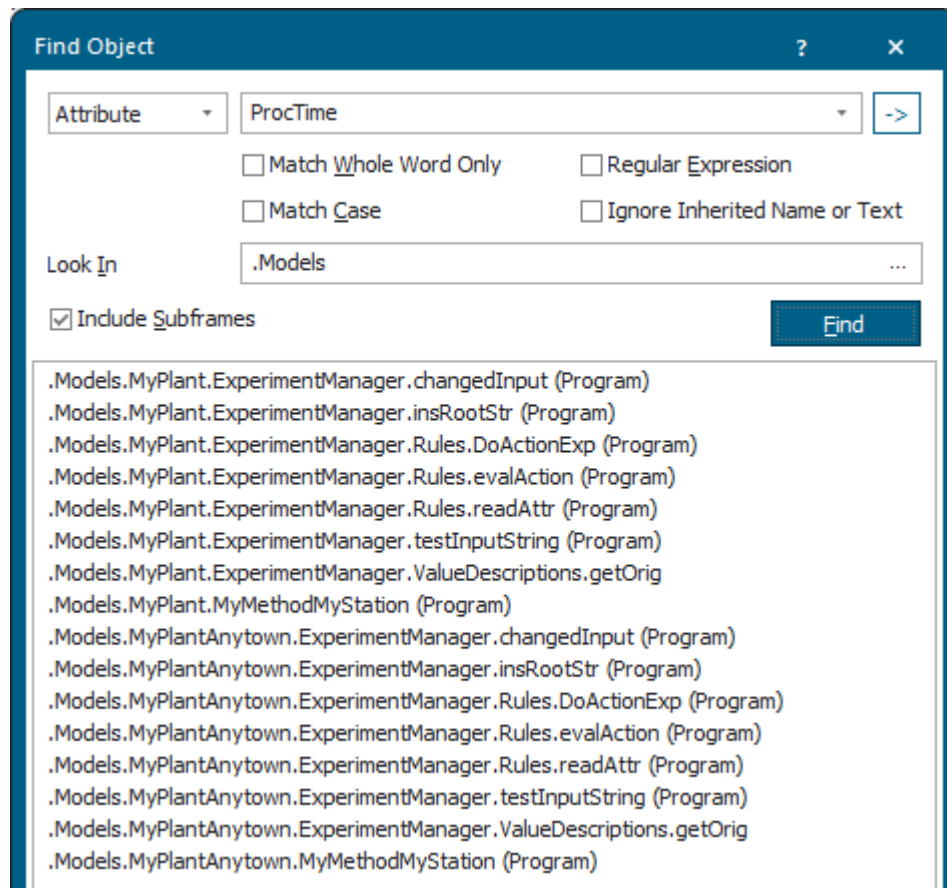
Back to [Find Objects and Text in Your Simulation Model](#)

Find Text in a Built-in or a User-defined Attribute of an Object

You can find any text or any value contained in a built-in or a user-defined attribute of an object which you inserted into your simulation model.

Proceed as follows:

1. Select **Attribute** from the left drop-down list.



2. Enter any text or value contained within a built-in or a user-defined attribute of an object in the search scope.

Plant Simulation shows the name of the attribute in which the text was found in parenthesis after the object in the results list.

3. Repeat steps 4 to 7 described under [Find the Name of an Object](#)
4. Click **Find**.

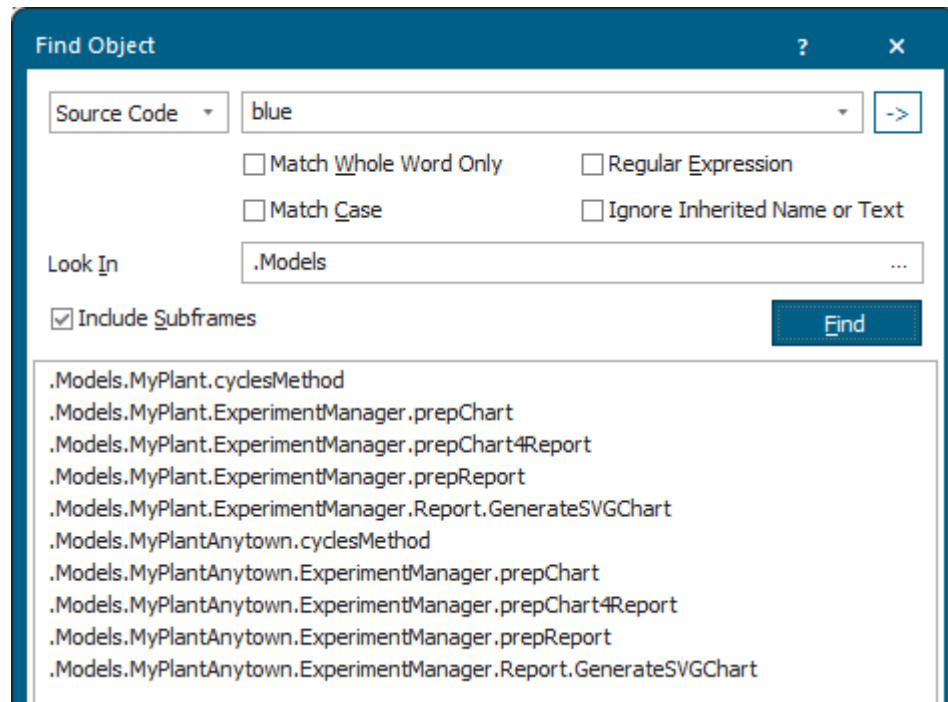
Back to [Find Objects and Text in Your Simulation Model](#)

Find Any Source Code in a Method

You can find any text, which you entered as source code into any *Method* object which you inserted into your simulation model or in any user-defined attribute of data type **method**.

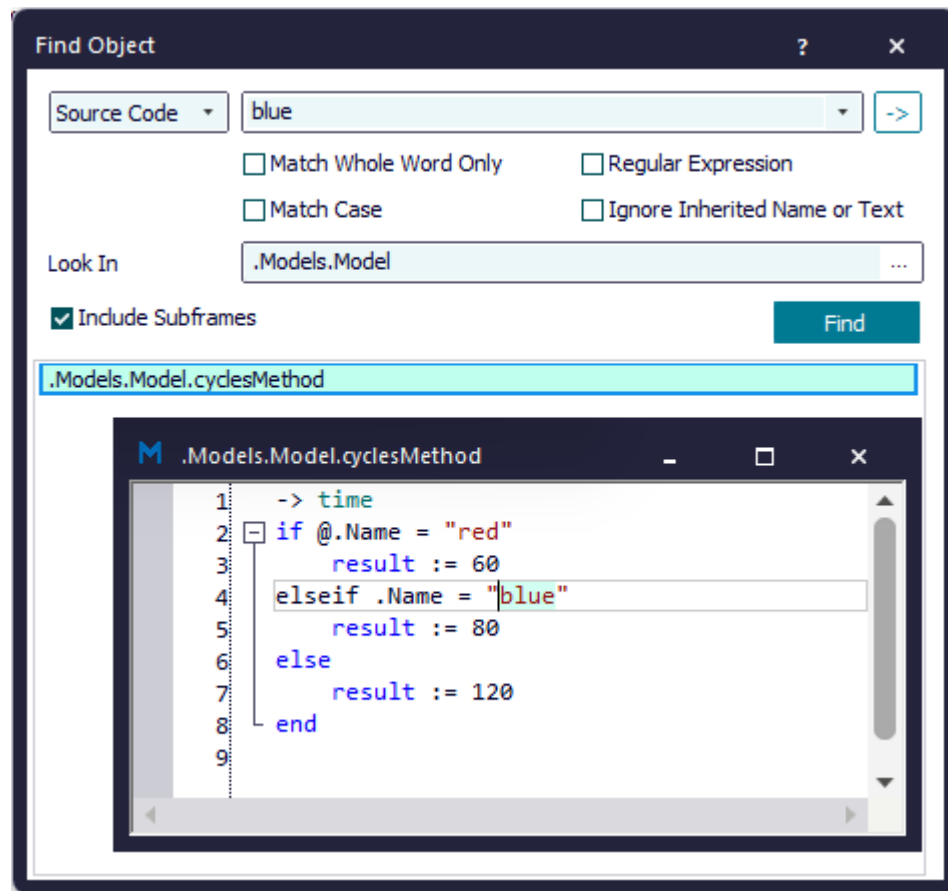
Proceed as follows:

1. Select **Source Code** from the left drop-down list.
2. Enter any text or value contained within the source code of a *Method* in the search scope.

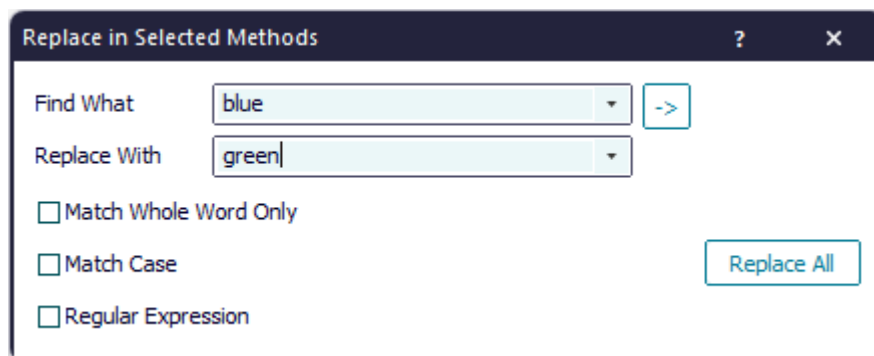


3. Repeat steps 4 to 10 described under [Find the Name of an Object](#).
4. When you select **Ignore Inherited Name or Text** *Plant Simulation* then only finds the original source code you entered into a *Method* object, and not the source code in derived or duplicated objects.
5. Click **Find**.
6. To open the *Method* that *Plant Simulation* found and to jump to the first occurrence of the item, double-click it in the results pane. Double-clicking or selecting **Open** again jumps to the next occurrence of the found item.

To jump to the previous occurrence in the source code, hold down **Shift** and double-click.



- To replace the found source code with other source code, click one or several *Methods* in the results list with the right mouse button and select the command **Replace With**. Then, enter the replacing source code into the dialog which opens.



Compare [Find Text \[in Method\]](#)

Back to [Find Objects and Text in Your Simulation Model](#)